Ho Man Colman Leung

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SUMMARY

PhD candidate building wearable health-sensing systems end to end, from custom sensors and circuit boards to signal processing, ML, and human-subject studies, including ECG in a loose T-shirt and a 363-day clinical glucose study. Seeking research scientist roles starting in 2027.

EDUCATION

Columbia University — New York, NY

PhD in Computer Science • Advisor: Prof. Xia Zhou — *2022 – May 2027 (expected)*

Began the PhD at Dartmouth College (2020 – 2022) and transferred to Columbia with my advisor's lab.

The Chinese University of Hong Kong — Hong Kong

MPhil in Computer Science and Engineering (2015 – 2018) • BSc in Computer Science (2009 – 2012)

EXPERIENCE

PhD Researcher, Mobile X Lab — 2020 – Present

Columbia University • Dartmouth College — *New York, NY • Hanover, NH*

Continuous ECG Monitoring with a Loose T-Shirt — *MobiDx 2026, MobiCom workshop (presenter)*

- Built a loose T-shirt that records ECG, a form factor long considered unusable due to motion artifacts, using a sweat-anchored kirigami sensor and a hybrid electrode that also captures motion.
- Developed a context-aware pipeline that uses the capacitive motion signal to adaptively filter ECG.
- Achieved 92.7% full-waveform detection at rest (r = 0.89 waveform correlation) and 84.5% R-peak detection while walking (4.58 bpm heart-rate error) in a pilot study.
- Participants preferred the shirt over commercial ECG patches and chest straps.

Noninvasive Optical Glucose Sensing — *Scientific Reports 2025 • Sensors 2023*

- Designed a portable sensor that estimates glucose levels from the rotation of polarized light, with a light-trap enclosure to reject reflections, and fabricated four prototypes.
- Led a 363-day home study of 30 participants with the Barbara Davis Center for Diabetes.
- Filtered out readings corrupted by improper device use and tuned Bayesian ridge regression models, with over 95% of predictions clinically accurate or acceptable.

Computational Fabrics (collaborations) — *IMWUT 2025 • ICORR 2025 • MobiSys 2024*

- Developed the Joey Android app, including integrating an OpenBCI dongle with the smartphone.
- Built the sleep-sensing pillowcase prototype and advised on a fabric hand-muscle sensor for rehabilitation.

Co-mentored 4 undergraduate students on lab research projects.

Research Associate — 2018 – 2020

The Chinese University of Hong Kong — *Hong Kong*

- Built a robust multi-camera, multi-object 6DoF real-time tracking system for medical simulators.
- Accelerated tracking with CUDA parallelization and pipelined execution to over 60 fps.

EXPERIENCE (CONTINUED)

MPhil Researcher 2015 – 2018
The Chinese University of Hong Kong *Hong Kong*

- Created TwistIn, which authenticates a smart device by matching its twist to the user's smartwatch motion.

Software Engineer 2012 – 2015
Dracaena Life Technologies Co., Ltd. *Hong Kong*

- Built medical training simulators, including a stereoscopic 3D anatomy tool deployed at a nursing school.

SELECTED PUBLICATIONS

Continuous Electrocardiogram Monitoring with a Loose T-Shirt [PDF]
Ho Man Colman Leung, Kyle Tong, Ming Gong, Nam Bui, Xia Zhou • *MobiDx 2026 (MobiCom workshop)*

Clinical Evaluation of a Polarization-based Optical Noninvasive Glucose Sensing System [PDF]
Ho Man Colman Leung, Chengyue Gong, Luke Geiser, Emily E. Fivekiller, Nam Bui, Tam Vu, Temiloluwa Prioleau, Gregory P. Forlenza, Qiang Liu, Xia Zhou • *Scientific Reports 2025*

My Pillow Knows My Sleep: Sleep Monitoring with Computational Fabrics in the Pillowcase [PDF]
Qijia Shao, Junxiao Chen, **Ho Man Colman Leung**, Meiqi Zhao, Ruoyu Xu, Jiting Liu, Lisa Maria DiSalvo García, Xiaofan Jiang, Marie-Pierre St-Onge, Xia Zhou • *IMWUT 2025*

Joey: Supporting Kangaroo Mother Care with Computational Fabrics [PDF]
Qijia Shao, Jiting Liu, Emily Bejerano, **Ho Man Colman Leung**, Jingping Nie, Xiaofan Jiang, Xia Zhou • *MobiSys 2024*

Noninvasive Glucose Sensing In Vivo [PDF]
Ho Man Colman Leung, Gregory P. Forlenza, Temiloluwa O. Prioleau, Xia Zhou • *Sensors 2023*

SKILLS

Hardware	Sensor design, e-textiles, PCBs (KiCad, EasyEDA), MCUs (STM32, ESP32), CAD, 3D printing
Data & ML	Biosignal processing, motion-artifact filtering, ML (PyTorch, scikit-learn, SciPy)
Study Design	Human-subject studies, from lab pilots to multi-month home deployments
Programming	Python, C/C++, C#, Java, MATLAB, CUDA, OpenGL, Unity, iOS, Android, Git, Linux
Languages	Cantonese (native), English (fluent), Mandarin

TEACHING

Teaching Assistant, Computer Networks 2021 – 2025
Columbia University (2024, 2025) • Dartmouth College (2021)

Teaching Assistant, Computer Graphics and Java, Final Year Project Tutor 2015 – 2017
The Chinese University of Hong Kong • Certificate of Merit

SERVICE & OUTREACH

Co-organizer & Technical Instructor, You(th)Fab Summer 2026
NYSCA-funded program teaching CAD, 3D printing and laser cutting to high school students

Volunteer Mentor, Columbia PhD Application Support 2023 – 2025
Feedback on statements and CVs for prospective PhD applicants

Reviewer, ACM IMWUT and ACM MobiSys